Week 4 – Performance & Load Testing

Performance Testing Report

Application: SauceDemo
Testing Tool: Apache JMeter

1. Objective

The objective of this performance test was to evaluate the responsiveness and stability of the SauceDemo web application when accessed by multiple concurrent users.

Apache JMeter was used to simulate virtual users and analyze the system's behavior under load conditions.

2. Load Test Scenario

The load test simulated multiple users accessing the SauceDemo homepage simultaneously.

Test Parameter	Value
Virtual Users	50
Ramp Up Time	10 seconds
Loop Count	5
Request Type	HTTP GET

Users were gradually introduced into the system to simulate real-world traffic conditions.

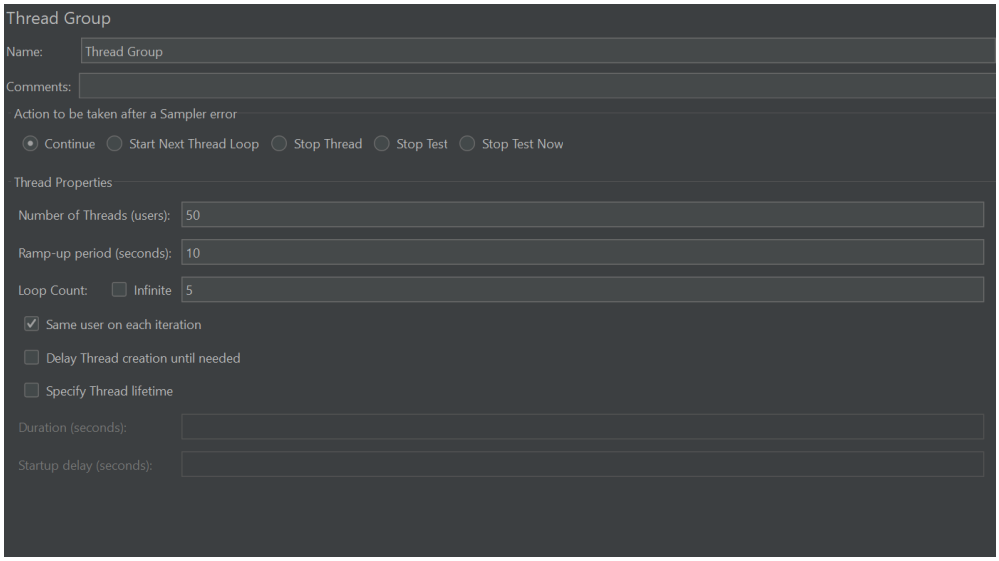

Figure 1: Thread Group Load Configuration

3. Observations

During the execution of the performance test, the system handled the simulated user load efficiently. The response time increased slightly as the number of concurrent users increased, which is expected during load testing.

The system remained stable throughout the testing process, and no major errors were observed.

The application responded successfully to user requests and maintained consistent performance.

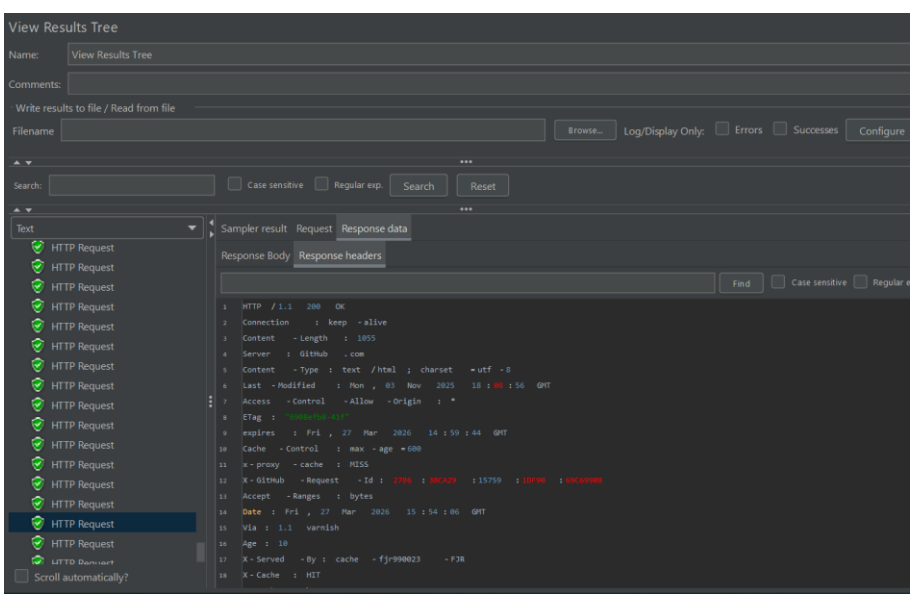

Figure 2: JMeter View Results Tree Output

4. Graph Results

The Graph Results listener in Apache JMeter displayed the response time behavior during the test execution.

The graph showed that response times gradually increased with the number of users but remained within acceptable limits.

The system did not show sudden spikes or performance degradation.

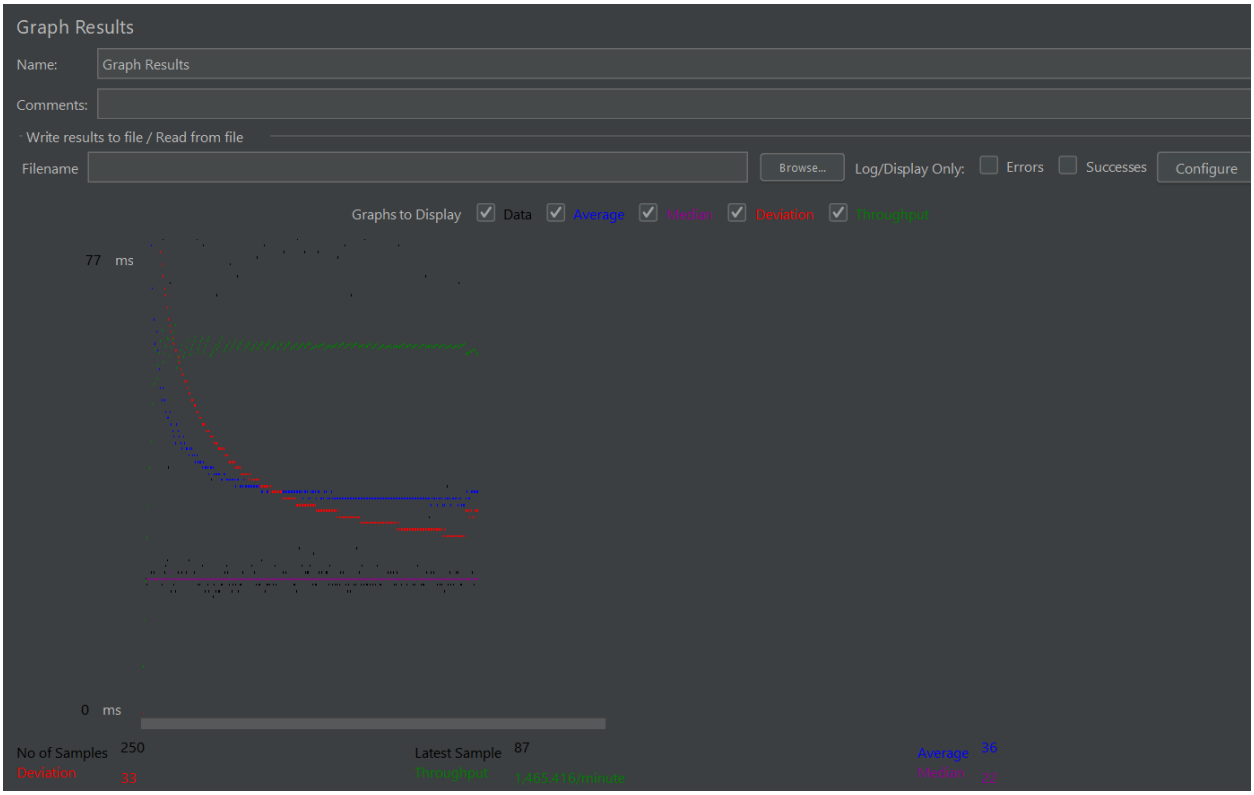

Figure 3: Response Time Graph from JMeter

5. Summary Report

The performance test generated a total of **250 samples**, representing the number of requests sent to the server during the test execution. The **average response time was 36 milliseconds**, indicating that the application responded quickly to user requests.

The **minimum response time recorded was 19 ms**, while the **maximum response time reached 308 ms** during peak load conditions. The **standard deviation of 33.2 ms** indicates moderate variation in response times during the test.

The **error rate remained at 0.00%**, meaning all requests were successfully processed by the server. The **throughput was 24.42 requests per second**, which represents the number of requests handled by the system per second during the test.

Additionally, the system received **40.49 KB of data per second** and sent **2.84 KB per second**, with an **average response size of 1697.6 bytes per request**.

Label	# Samples	Average Response Time (ms)	Min (ms)	Max (ms)	Std. Dev.	Error %	Thoughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	250	36	19	308	33.2	0.00%	24.42 requests/sec	40.49	2.84	1697.6
TOTAL	250	36	19	308	33.2	0.00%	24.42 requests/sec	40.49	2.84	1697.6

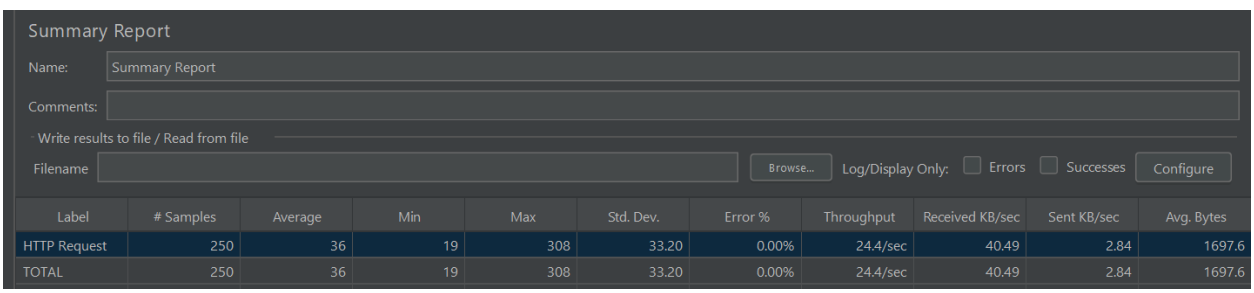

Figure 4: JMeter Summary Report

6. Conclusion

The performance testing results indicate that the SauceDemo application performs efficiently under moderate load conditions. The response times remained within acceptable limits, and the application handled concurrent user requests without failures.

The test confirms that the application is capable of supporting multiple users while maintaining stable performance.

7. Recommendations

The following improvements are recommended to further enhance system performance:

- Optimize server-side processing
- Implement caching mechanisms
- Conduct stress testing with higher user loads
- Use load balancing techniques for high traffic scenarios